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Major Project Report
on
DocsGPT: AI-Powered Document Question Answering System
MCA491P

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Submitted in partial fulfilment for the award of degree of
MASTER OF COMPUTER APPLICATIONS
2025-26



CERTIFICATE

Certified that the project work titled "**DocsGPT: AI-Powered Document Question Answering System**", carried out by Pavan Mahantesh Kurubar, USN: 1RV24MC072, a bonafide student, submitted in partial fulfilment for the award of Master of Computer Applications of RV College of Engineering®, Bengaluru, affiliated to Visvesvaraya Technological University, Belagavi, during the year 2025-26. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report deposited in the departmental library. The project report has been approved as it satisfies the academic requirement in respect of project work prescribed for the said degree.

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ABSTRACT

The rapid growth of digital documents in educational, research, and enterprise environments has created significant challenges in efficiently locating and retrieving relevant information. Traditional document search methods primarily rely on manual reading and keyword-based retrieval, which are often time-consuming and may fail to capture the contextual meaning of user queries. This limitation increases the effort required to analyze large volumes of documents and extract useful knowledge. Recent advancements in Artificial Intelligence (AI), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and semantic search techniques have enabled the development of intelligent systems capable of understanding document content and providing context-aware responses. However, many existing solutions focus only on basic document retrieval or chatbot functionalities and do not offer a complete framework for intelligent document analysis and question answering.

The proposed system, titled **“DocsGPT: AI-Powered Document Question Answering System,”** is designed to address these challenges by providing an intelligent platform for document understanding and information retrieval. The system allows users to upload documents such as PDFs, reports, research papers, and scanned documents, and interact with them using natural language queries. The architecture integrates multiple components including document upload and processing, Optical Character Recognition (OCR), semantic search, vector embedding generation, Retrieval-Augmented Generation (RAG), and Large Language Models (LLMs). Uploaded documents are processed, converted into vector embeddings, and stored within a vector database to enable efficient semantic retrieval. When a user submits a query, the system retrieves the most relevant document content and uses an LLM to generate accurate, context-aware responses based on the retrieved information.

The implementation utilizes modern technologies such as Flask for backend services, React and Vite for frontend development, MongoDB/PostgreSQL for data storage, Redis for background task processing, and FAISS for vector similarity search. OCR capabilities are incorporated to support scanned and image-based documents, ensuring broader document compatibility. The system provides a user-friendly interface that simplifies document interaction while maintaining scalability and performance. Experimental evaluation demonstrates that the proposed solution effectively improves document retrieval accuracy, reduces manual searching effort, and delivers meaningful responses with high contextual relevance. The project highlights the practical application of AI-driven document analysis and knowledge retrieval, making it valuable for educational institutions, research organizations, and enterprise knowledge management systems.

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TABLE OF CONTENTS

Contents	Page No.
College Certificate	i
Declaration	ii
Acknowledgement	iii
Abstract	iv
List of Publications	v
Table of Contents	vi
List of Tables	viii
List of Figures	ix
Chapter 1: Introduction	1 - 3
1.1 Project Description	1
1.2 Company Profile	2
1.3 Dissertation Organization	3
Chapter 2: Literature Review	4 - 9
2.1 Literature Survey	4
2.2 Existing and Proposed System	5
2.3 Tools and Technologies Used	7
2.4 Hardware and Software Requirements	8
Chapter 3: Software Requirement Specifications	10 - 14
3.1 Introduction	10
3.2 General Description	11
3.3 Functional Requirements	12
3.4 External Interface Requirements	13
3.5 Non-Functional Requirements	13
3.6 Design Constraints	14
Chapter 4: System Design	15 - 18
4.1 Architectural Design	15
4.2 Context Diagram	17
Chapter 5: Detailed Design	19 - 30

Contents	Page No.
5.1 Class Diagram	19
5.2 Use Case Diagram	21
5.3 Activity Diagram	22
5.4 Sequence Diagram	23
5.5 Data Flow Diagrams	24
5.6 Database Design	26
5.7 Module-wise Logic Description (PDL)	27
Chapter 6: Implementation	31 - 36
6.1 Code Snippets	32
6.2 Implementation Details	34
Chapter 7: Software Testing	37 - 42
7.1 Test Cases	37
7.2 Testing and Validations	41
Chapter 8: Conclusion	43 - 44
Chapter 9: Future Enhancements	45 - 46
Bibliography	47 - 50

LIST OF TABLES

Table No.	Table Name	Page No.
7.1	Functional Test Cases	42
7.2	Integration Test Cases	43
7.3	System Test Cases	44
7.4	User Acceptance Test Cases	45

LIST OF FIGURES

Fig No.	Figure Name	Page No.
4.1	Overall System Architecture	17
4.2	Context Diagram – User and External Services	19
5.1	Use Case Diagram	21
5.2	Activity Diagram – Document Upload and Query Flow	22
5.3	Sequence Diagram – User Query Processing	23
5.4	DFD Level 0	24
5.5	DFD Level 1	25
5.6	DocsGPT Home Interface	26
5.7	Document Upload Interface	27
6.1	Agent Creation Interface	36
6.2	Chat Interface with AI Response	37